

Dissipation and movement of soil water in artificial forest in arid oasis areas: Cognition based on stable isotopes

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ABSTRACT

Large-scale construction of artificial forests in arid oasis areas plays a role in wind prevention and sand fixation, vegetation increase and landscape improvement. However, due to the extremely strong evapotranspiration in the arid areas, the construction of artificial forests may greatly increase the consumption of water resources. Therefore, it is necessary to evaluate and weigh the benefits of its ecology, landscape and water resource utilization. Based on different water isotopic data (soil water, precipitation) and related hydrometeorological data of typical poplar plantation in the middle and lower reaches of Shiyang River Basin from 2018 to 2019, this study evaluated the soil evaporation loss rate and water storage capacity of artificial forest and revealed the mechanism of soil water movement. The results showed that: (1) The average soil evaporation loss in the artificial forest was 21.30%, which was higher in spring and summer and lower in autumn and winter; (2) The soil water recharge from the artificial forest has the characteristics of preferential flow recharge, with long retention of precipitation in the soil, and full mixing of newly recharged water with original water; (3) The artificial forest increased the water storage capacity of deep soils (80–100 cm) in the oasis area, but the water consumption rate also correspondingly increased. Artificial forests must rely on water input to maintain normal growth, and its own water consumption rate and soil evaporation loss are high. Therefore, ecological, landscape, and water resource utilization benefits should be considered when constructing artificial forests in arid oasis areas where water resources are scarce.

1. Introduction

Soil moisture is a key link to the water transformation of each interface in the water cycle of the groundwater-soil-plant-atmosphere continuum (GSPAC system for short) (Philip, 1966), and an important factor affecting forest water conservation, soil retention function and vegetation restoration (Shen et al., 2019). As an important part of the water cycle, soil water in unsaturated areas can be converted from precipitation to river or groundwater recharge. Identification of soil water evaporation, infiltration and storage characteristics is critical to understanding regional hydrological cycles and water balance under environmental changes (Brooks et al., 2010; Dubbert and Werner, 2019).

The water cycle and water balance in an area is controlled by precipitation, evapotranspiration (ET), runoff, infiltration and soil water

storage (Bonan, 2016). In arid areas, plantation construction can increase water vapor in the air through transpiration and thus increase precipitation to some extent, but it should be noted that precipitation is also affected by other factors, such as atmospheric circulation, so further analysis is needed (Li et al., 2018). Artificial forests can also reduce runoff loss and regulate seasonal runoff distribution through precipitation interception and accumulation (Feng et al., 2016). In addition, a certain thickness of litter accumulates on the forest's surface, and there is well-structured soil and a developed root system underneath, so constructing an artificial forest can increase soil water infiltration and water conservation ability (Liu et al., 2019a). There are two movement modes of soil water infiltration, namely preferential flow and plug flow. Plug flow is the complete mixing of water through the soil matrix with shallow free water. Under the action of plug flow, precipitation infiltrates along the hydraulic gradient, pushing the original soil water

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downward. Preferential flow means that precipitation bypasses the soil matrix and quickly enters deep soil by using soil macropores and rotten roots (Tan et al., 2017; Zheng et al., 2019). As for the impact of artificial forest on water storage in forest areas, studies show that artificial forest in arid areas intensifies soil water consumption and reduces soil water storage (He et al., 2003; McVicar et al., 2007). Present problems concerning water cycle and water balance in arid areas include: previous studies focus on the decline of soil moisture and runoff due to increased ET in arid areas (He et al., 2003; Feng et al., 2016); however, the response of precipitation to vegetation increase is ignored, and the research shows that vegetation increases to a certain extent can offset some negative effects of ET increase on soil moisture (Li et al., 2018). In addition, few studies address interactions among precipitation, evapotranspiration, soil water movement, and infiltration storage of artificial forests in arid areas.

Hydrogen and oxygen stable isotope ($\delta^2\text{H}$, $\delta^{18}\text{O}$) are two natural tracers of water movement and have been widely used in tracing research of ecohydrological processes such as evaporation (Zhu et al., 2021a), soil water movement process (Zheng et al., 2019), and groundwater recharge (Duvert et al., 2016). Water infiltration and evaporation in unsaturated soil are the main forms of soil water movement. Changes in soil water isotopes and soil water at a certain soil depth can identify seepage methods of precipitation (Zhu et al., 2022a). Moreover, the dynamic water process reflected by the displacement of the soil water isotope signal is also called the “memory effect”. Understanding the “memory effect” will help us track the dynamics of climate and soil hydrology (Kleine et al., 2020). Since water uptake by plant roots does not cause isotope fractionation of soil water isotope (Allison and Barnes 1983; Dawson and Ehleringer 1991), soil evaporation is generally considered as the main factor causing isotopic enrichment of infiltrated precipitation in soil. However, soil evaporation or evaporation peak usually occurs within 0–30 cm of the surface layer (Sprenger et al. 2016), and evaporation decreases exponentially with the increase of soil depth (Allison and Barnes, 1983). In addition, changes of soil water isotope signals at a certain soil depth can identify water infiltration methods (Peralta-Tapia et al., 2015). Among them, preferential inflow infiltration shows high variability in isotopic signals (Brodersen et al., 2000). After evaporation and infiltration, some of the precipitation will be stored in the soil.

In the arid areas of northwest China, the main way to improve the ecological environment is to build artificial forests (He et al., 2021). Artificial forests can prevent desertification, improve the site environment and enrich species diversity in arid areas, thus playing an important role in the forest ecosystem (Zastrow Mark, 2019). Poplar becomes one of the main introduced plantation species in the arid areas of northwest China due to its rapid growth and drought resistance (Liu et al., 2018). Therefore, in this study, Minqin Oasis, where water resources are scarce and ecological environment is fragile, was selected as the research area. The stable isotopes of water (soil water and precipitation) in poplar plantation were regularly monitored at fixed point for two consecutive years, and the comparative observation of soil moisture with natural landscape such as shrubs and grasslands was designed. In order to better understand the mechanism of soil water dissipation and movement in arid artificial forest and achieve the maximum benefit of artificial forest in arid oasis area, we proposed the following research objectives: (1) quantify the soil evaporation loss rate and water storage capacity of poplar plantations in arid areas; (2) reveal the mechanism of soil water movement in poplar plantation in arid areas. This study systematically analyzed the dissipation and movement of soil water input in artificial forest, and the results could provide a scientific basis for coordinating ecological benefits, landscape benefits and water resource utilization benefits in arid areas.

2. Materials and methods

2.1. Site description

The study area is located in Hongyashan Branch of Shiyang River Forest Farm. Hongyashan Forest Farm is located in the middle and lower reaches of Shiyang River basin, with a geographic location of 102° 54' E, 38° 24' N (Fig. 1a) and a total area of 51313 ha. The study area belongs to the temperate continental arid climate, with an average altitude of 1400 m, an average annual temperature of 9.2 °C, a frost-free period of 210 days. Annual average rainfall is 187.9 mm and is mainly concentrated between June and September (Zhu et al., 2022b). The main vegetation types in Hongyashan Forest Farm include *Populus tomentosa* Carr., *Nitraria tangutorum* Bobr., *Calligonum mongolicum* Turcz., *Haloxylon ammodendron* (C. A. Mey.) Bunge, *Kalidium foliatum* (Pall.) Moq., and *Artemisia desertorum* Spreng., etc., and the existing artificial forest area is 17600 ha.

2.2. Sampling design and sample collection

The sampling site is located in Hongyashan Forest Farm in Minqin Oasis. In the monitored forest, we selected four 10 × 10 m plots (2 plantation plots, 1 shrub plot and 1 grassland plot) with similar habitats and the same age. Then, two 1 × 1 m quadrats were randomly selected in each of the four plots. Finally, eight quadrats (4 in artificial forests, 2 in shrubs and 2 in grasslands) were set up. We excavated a 1 m deep soil profile within each quadrat to collect soil samples. All samples in this study were collected once a month during the plant growth period (April–October) from 2018 to 2019.

2.2.1. Precipitation sampling

We placed a China standard precipitation gauge (CSPG, height 70 cm, diameter 20 cm, resolution 0.1 mm) near the sampling point, installed a funnel (diameter 20 cm) above the precipitation gauge to collect rainwater, and placed a polyethylene anti-evaporation bulb at the funnel mouth to prevent isotope fractionation caused by evaporation. Corresponding precipitation samples were collected by local farmers (after passing a professional training, they were entrusted to collect samples). In order to avoid experimental errors caused by sample evaporation, samples were immediately collected and placed into 100 ml high-density polyethylene vials after each precipitation. Three parallel samples were taken each time, sealed with Parafilm film, and refrigerated (4 °C) until experimental analysis. During the observation period (April–October 2018 and 2019), we collected 132 precipitation samples.

2.2.2. Soil sampling

A soil profile of 100 cm was excavated at the selected sampling point, and soil samples were collected at intervals of 10 cm. The collected samples were mainly divided into two parts. One part was undisturbed soil samples taken with a ring knife in layers at 10 cm intervals. One soil samples were taken from each layer and brought back to the laboratory for soil bulk density and moisture measurement. The other parts were put into a 50 ml glass bottle, sealed and marked with the sampling date, and then frozen (−20 °C) to determine the hydrogen and oxygen isotopes of soil water. Three parallel samples were also taken from each layer. During the observation period (April–October 2018 and 2019), we collected samples once a month and collected a total of 1120 soil samples.

2.2.3. Meteorological data collection

We used air temperature, relative humidity and precipitation to indicate the temperature and humidity conditions at the observation site. Meteorological data, like temperature and relative humidity were obtained through long-term continuous meteorological observation by the automatic meteorological station (WatchDog 2000 series) set up

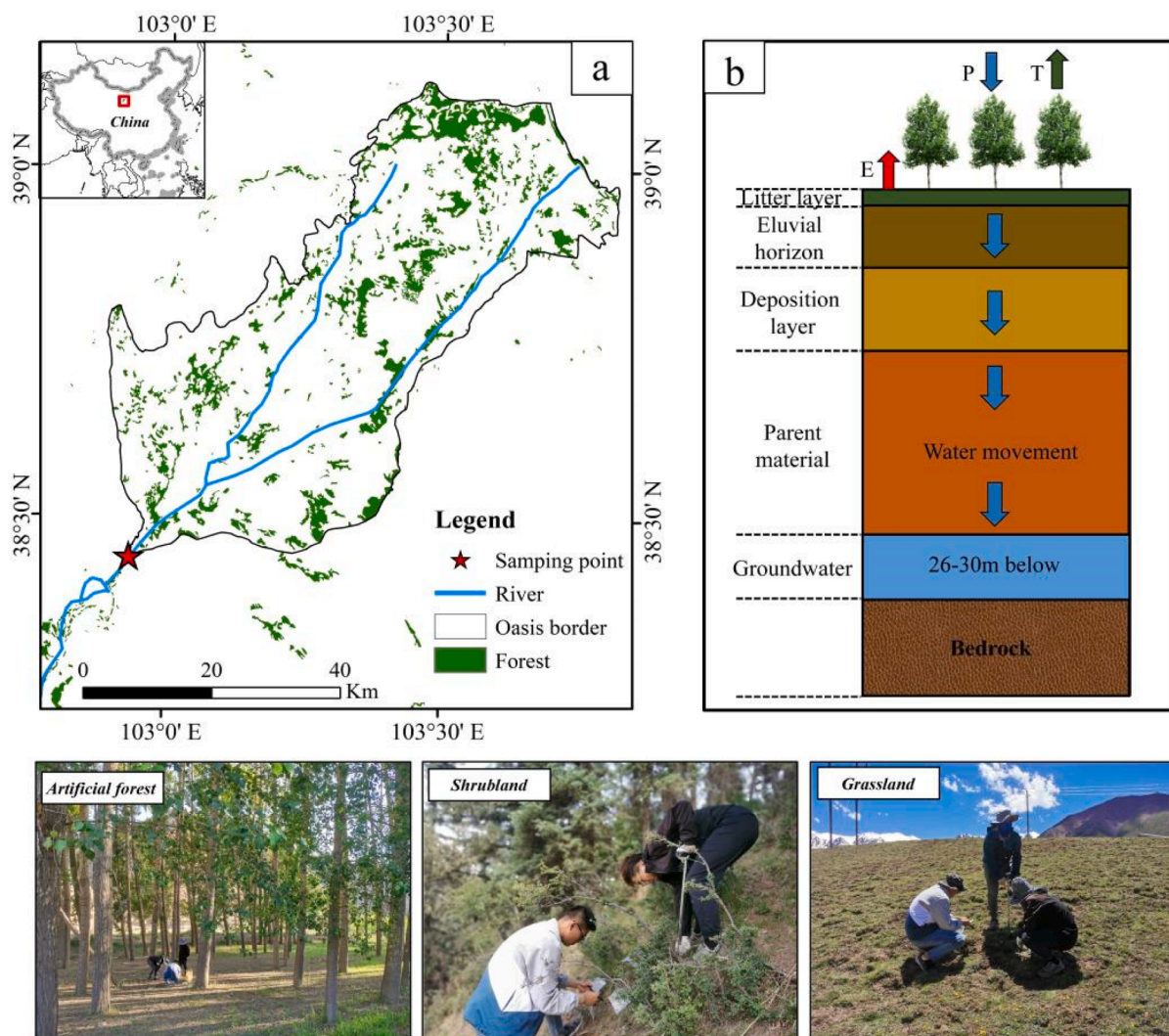


Fig. 1. Overview of the study area.

near the sampling point. Since the precipitation data collected by the local farmer may have some errors, we compared them with the daily precipitation data provided by the China Meteorological Data Network (<https://data.cma.cn/>). We found that the precipitation data of Wuwei Station can better represent the precipitation in Hongyashan Forest Farm. Therefore, for the temperature, relative humidity and other data in this study, we used the measured data, while for the precipitation data, we used the data provided by Wuwei Station (China Meteorological Data Network) from 2018 to 2019.

2.3. Methods

2.3.1. Sample analysis

All the analyses in this study were conducted in the Stable Isotope Laboratory of the College of Geography and Environmental Science, Northwest Normal University, China. The frozen soil samples were gradually thawed at 0–4 °C, and then water was extracted from the soil using an Automatic Cryogenic Vacuum Distillation Water Extraction System (LI-2100, Beijing LICA United Technology Limited, China). After extraction, the extracted water (solid water) was first melted at room temperature, then filtered through a syringe membrane filter with 0.45 μm pore diameter and 13 mm diameter by a syringe and lastly put into a 2 ml vial. To avoid cross contamination, each syringe was only used for just one sample, and each vial was injected with 0.5–1.5 ml sample to keep the sample volume stable. After that, the cap with the diaphragm

was immediately screwed to avoid isotope fractionation. Hydrogen and oxygen stable isotopes of all water samples (including precipitation and soil water) were determined using a liquid water isotope analyzer (DLT-100, Los Gatos Research, Inc., Mountain View, CA, USA). Each water sample and isotope standard sample were injected 6 times continually during the analysis. In order to avoid the memory effect of isotope analysis, the values of the first two injections were discarded, and the average value of the last 4 times was used as the final isotope value (Zhu et al., 2021a, 2021b). Sample analysis results were expressed in thousandths difference relative to the Vienna Standard Mean Ocean Water (V-SMOW) as follows:

$$\delta = \left[\left(\frac{R_{\text{sample}}}{R_{\text{standard}}} \right) - 1 \right] \times 1000\text{‰} \quad (1)$$

Where R_{sample} represents the ratio of $^{18}\text{O}/^{16}\text{O}$ or $^2\text{H}/^1\text{H}$ in the sample, and R_{standard} is the ratio of $^{18}\text{O}/^{16}\text{O}$ or $^2\text{H}/^1\text{H}$ in V-SMOW. The test error of $\delta^2\text{H}$ value was not more than $\pm 0.6\text{‰}$, and the test error of $\delta^{18}\text{O}$ value was not more than $\pm 0.2\text{‰}$.

2.3.2. Lc-excess

In this study, the soil evaporation intensity was determined qualitatively by Lc-excess. The linear relationship between precipitation $\delta^2\text{H}$ and $\delta^{18}\text{O}$ and soil moisture are the local meteoric water line (LMWL) and soil water line (SWL), respectively. The linear correlation lines between

$\delta^2\text{H}$ and $\delta^{18}\text{O}$ of atmospheric precipitation and soil water in a specific region are LMWL and SWL, respectively. The linear offset (Lc-excess) of different water bodies relative to the local meteoric precipitation line $\delta^2\text{H}$ proposed by Landwehr and Coplen is an indicator of evaporation degree (Landwehr and Coplen, 2004):

$$\text{Lc - excess} = \delta^2\text{H} - a\delta^{18}\text{O} - b \quad (2)$$

Where a and b are the slope and intercept of LMWL, respectively, and $\delta^2\text{H}$ and $\delta^{18}\text{O}$ are the hydrogen and oxygen isotope values in the sample, respectively. When the $\delta^2\text{H}$ - $\delta^{18}\text{O}$ line of the water body is lower than LMWL and the Lc-excess is less than 0, it indicates that the stable isotopes in the water body are enriched due to evaporation. When the water body Lc-excess has a positive value, it indicates that the water sample may be affected by other water sources, besides precipitation (Landwehr et al., 2014).

2.3.3. Craig-Gordon model

In this study, the Craig-Gordon model quantitatively calculated the soil evaporation loss rate. Based on the principle of isotope mass balance, evaporation loss in an open liquid-vapour isotope system can be calculated by the Craig-Gordon model (Gonfiantini, 1986) through the following formula:

$$f = 1 - \left[\frac{(\delta_s - \delta^*)}{(\delta_p - \delta^*)} \right]^m \quad (3)$$

Where: δ_s is defined as the average isotope value of soil water at different depths; δ_p is defined as the isotopic composition of the original water source, which can be calculated by the intercept of the intersection of LMWL and SEL in the open liquid-vapour isotope system (Evaristo et al., 2015). The formula is as follows:

$$\delta^{18}\text{O}_{\text{intercept}} = \frac{x - b}{a - y} \quad (4)$$

$$\delta^2\text{H}_{\text{intercept}} = a\delta^{18}\text{O}_{\text{intercept}} + b \quad (5)$$

Where a and b , x and y are the slope and intercept of LMWL and SEL, respectively. The slope of the local evaporation line S_{SEL} can be calculated by the following formula (Gonfiantini, 1986):

$$x = S_{\text{SEL}} = \frac{\left[\frac{h(\delta_{\text{rain}} - \delta_A) - (1 + \delta_{\text{rain}} \times 10^{-3})(\epsilon_k + \epsilon^+/\alpha^+)}{1 - h + \epsilon_k \times 10^{-3}} \right]^{2\text{H}}}{\left[\frac{h(\delta_{\text{rain}} - \delta_A) - (1 + \delta_{\text{rain}} \times 10^{-3})(\epsilon_k + \epsilon^+/\alpha^+)}{1 - h + \epsilon_k \times 10^{-3}} \right]^{18\text{O}}} \quad (6)$$

Based on soil moisture isotopes ($\delta^2\text{H}_s$ and $\delta^{18}\text{O}_s$), the intercept of the soil evaporation line (I_{SEL}) can be calculated as:

$$y = I_{\text{SEL}} = \delta^2\text{H}_s - S_{\text{SEL}} \times \delta^{18}\text{O}_s \quad (7)$$

δ^* is the limiting factor of isotope enrichment (Gonfiantini, 1986), which the following formula can calculate:

$$\delta^* = \frac{h\delta_A - \epsilon_k + \epsilon^+/\alpha^+}{h - 10^{-3}(\epsilon_k + \epsilon^+/\alpha^+)} \quad (8)$$

Where: h is the relative humidity in the air, ϵ_k is the kinetic fractionation coefficient, and the formulas for $\delta^2\text{H}$ and $\delta^{18}\text{O}$ are as follows (Merlivat, 1978):

$$\epsilon_k(^{2\text{H}}) = n \times (1 - h) \times (1 - 0.9755) \quad (9)$$

$$\epsilon_k(^{18\text{O}}) = n \times (1 - h) \times (1 - 0.9723) \quad (10)$$

Where n is the aerodynamic parameter of the liquid-air interface during evaporation. Generally, 0.50 is for saturated soil and 1 for dry soil

(Benettin et al., 2018). δ_A is the isotope value in the surrounding air and is calculated as:

$$\delta_A = \frac{\delta_{\text{rain}} - k\epsilon^+}{(1 + k\alpha^+ \times 10^{-3})} \quad (11)$$

Where δ_{Rain} is the isotope value in precipitation, ϵ^+ and α^+ is the equilibrium fractionation coefficient (Horita and Wesolowski, 1994), and the formula is as follows:

$$\epsilon^+ = (\alpha^+ - 1) \times 1000 \quad (12)$$

$$10^3 \ln[\alpha^+(^2\text{H})] = 1158.8(T^3/10^9) - 1620.1(T^2/10^6) + 794.84(T/10^3) - 161.04 + 2.9992(10^9/T^3)$$

$$10^3 \ln[\alpha^+(^{18}\text{O})] = -7.685 + 6.7123(10^3/T) - 1.6664(10^6/T^2) + 0.3504(10^9/T^3)$$

Where T represents air temperature (K). M is the slope of enrichment (Gibson and Reid, 2014), and the calculation formula is as follows:

$$m = \frac{h - 10^{-3}(\epsilon_k + \epsilon^+/\alpha^+)}{1 - h + 10^{-3}\epsilon_k} \quad (15)$$

2.3.4. Soil water storage and water consumption rate

Soil water storage refers to the water layer thickness formed by all water in a soil layer (Milly, 1994), which can be expressed as follows:

$$S = R \times W \times H \times 10 \quad (16)$$

Where S is the soil water storage of a certain thickness layer (mm), R is the soil bulk density (g/cm³), and H is the soil thickness (cm). The following formula expresses mass water content W :

$$W = \frac{M_1 - M_2}{M_2} \times 100\% \quad (17)$$

Where M_1 is the weight value of wet soil (g) and M_2 is the weight value of dry soil (g).

Water consumption rate V_s is the variation of soil water storage in unit time (mm/a), and the calculation formula is:

$$V_s = \frac{\Delta S}{T} \quad (18)$$

Where ΔS is the difference (mm) between the initial soil water storage and the present soil water storage. In this study, the water storage of the control lands (grassland, shrub) are regarded as the initial soil water storage in this region to compare the water consumption rate of the artificial forest, grassland and shrub. T is the number of years corresponding to the change of soil water storage (a).

2.3.5. Pearson correlation coefficient

Pearson correlation coefficient is a statistical indicator that reflects the degree and direction of linear correlation between two variables. The value interval is $[-1, 1]$, where 0 indicates that the two variables are not correlated; positive value indicates positive correlation; negative value indicates negative correlation; and the larger the absolute value is, the stronger the correlation is. The calculation formula is as follows (Pearson, 1909):

$$r_p = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 (y_i - \bar{y})^2}} \quad (19)$$

Where, r_p stands for Pearson correlation coefficient; x_i and y_i are the

observed sample values of the two variables; $\bar{x}$ and $\bar{y}$ represent the average value of the two, respectively; n is the number of sample observations. In addition, correlation is influenced by significance, with significance less than 0.05 being significantly correlated and less than 0.01 being extremely significant.

3. Results

3.1. Soil water evaporation loss in arid artificial forest

3.1.1. Local meteorological conditions

Local meteorological conditions affected the soil water evaporation loss in the artificial forest. During the observation period from 2018 to 2019, the daily average temperature was 16.37 °C, ranging from −1.53 °C (April 5, 2018) to 28.3 °C (July 17, 2018). The daily average relative humidity was 53.14%, ranging from 14.25% (April 1, 2018) to 94.5% (September 9, 2019). The annual average precipitation was 221.47 mm, mainly concentrated from June to August (Fig. 2a,c). Affected by complex weather conditions during precipitation, the isotopic composition of precipitation in the study area was different during the observation

period. The mean $\delta^2\text{H}$ and $\delta^{18}\text{O}$ values of atmospheric precipitation were -38.32 ± 20.63 ‰ (-72.79 ‰ $\sim$ -4.49 ‰) and -5.96 ± 2.89 ‰ (-10.44 ‰ $\sim$ -0.81 ‰), respectively, in 2018, and -27.91 ± 24.28 ‰ (-73.92 ‰ $\sim$ 23.45 ‰) and -4.80 ± 3.39 ‰ (-10.23 ‰ $\sim$ 1.73 ‰), respectively, in 2019. Isotopes of atmospheric precipitation in the study area during 2018–2019 had similar temporal variations, with fluctuations of $\delta^2\text{H}$ and $\delta^{18}\text{O}$ increasing from April to July, reaching a maximum in July and then gradually decreasing (Fig. 2b,d).

3.1.2. Soil water evaporation loss

Soil evaporation loss f represents the percentage of soil water lost in the process of precipitation infiltration to evaporation. Compared with lc -excess, evaporation loss f can quantitatively describe the strength of soil evaporation. During the observation period from 2018 to 2019, the average evaporation loss f was 21.30% (Fig. 3a), showing a trend of increasing first and then decreasing in time variation. From April to July 2018, evaporation loss f gradually increased, and the monthly average evaporation loss was 16.82%, 15.10%, 21.73% and 32.64%, respectively. In August, it reached the maximum in the year, with an average of 33.62%. Then it decreased gradually. In September, the average

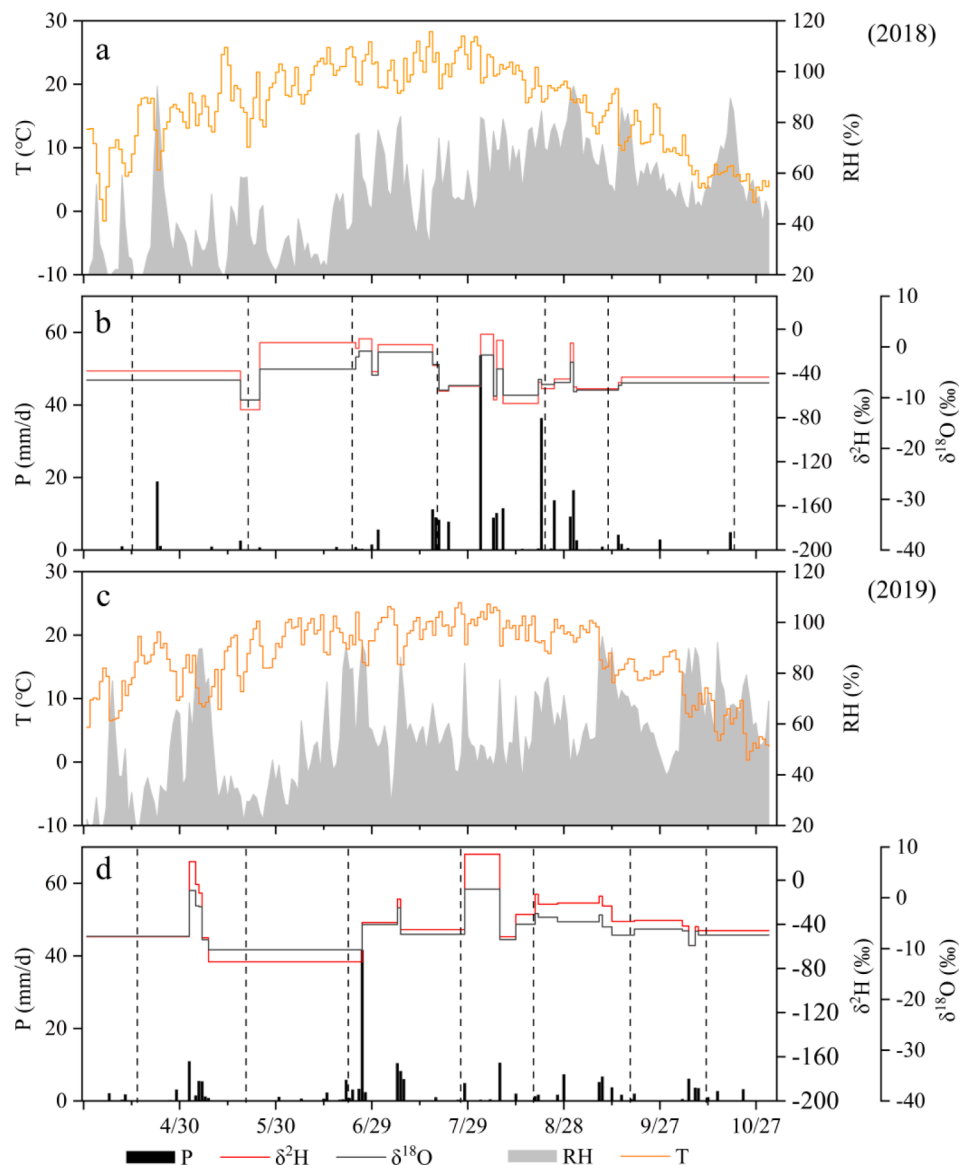


Fig. 2. Time series of meteorological and hydrological conditions and precipitation isotopes in the study area (a, c. temperature and relative humidity data for 2018 and 2019, respectively; b, d. precipitation and precipitation stable isotope data for 2018 and 2019, respectively; dotted line represents sampling time of soil water).

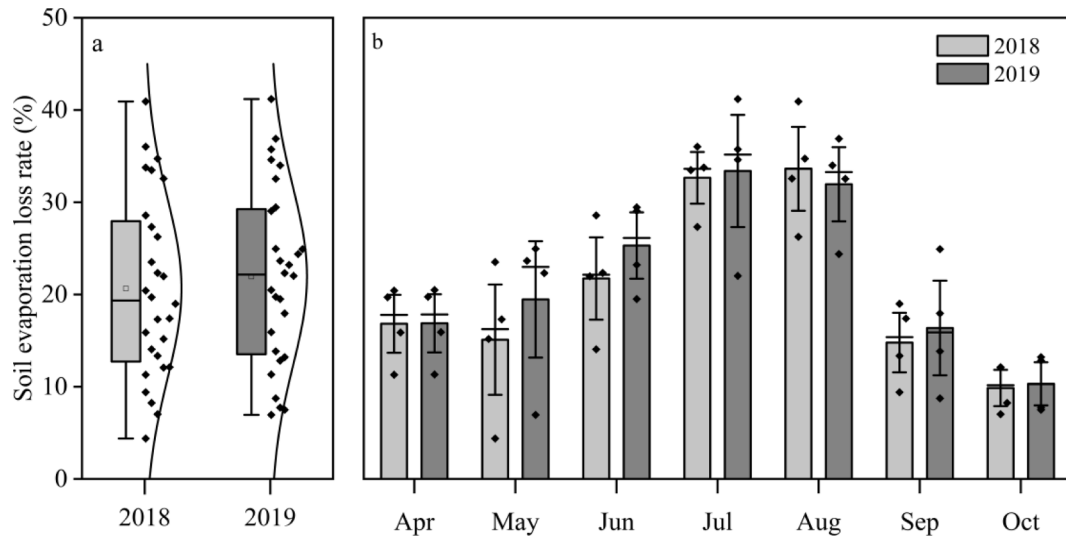


Fig. 3. Soil water evaporation loss rate f in artificial forest (a. Annual variation of soil evaporation loss rate f ; b. Monthly variation of soil water evaporation loss rate f).

evaporation loss f reduced to 14.79%, and then to 9.86% in October, which was the lowest value during the observation period. The variation law of evaporation loss f in 2019 was basically consistent with that of the previous year, except that the average evaporation loss in July was the largest, which was 33.39% (Fig. 3b).

3.1.3. Factors of soil evaporation in arid artificial forest

Soil evaporation was caused by a variety of factors. When expressing evaporation loss, f also includes many other factors affecting soil evaporation, so we analyzed the correlation between f and other factors (Fig. 4). f was positively correlated with average temperature and precipitation, and the correlation coefficients were 0.78 and 0.55, respectively. Since the precipitation was concentrated from June to August and the average temperature was the highest, the soil evaporation loss rate from June to August was also higher than that in other months. f was

negatively correlated with the relative humidity, with a correlation coefficient of -0.23 , which indicated that the higher the humidity, the weaker the evaporation was, but the correlation was also low. In addition, in soil evaporation, soil water storage and SWC represented the amount of evaporation water in the soil, so they were also important factors affecting soil evaporation intensity. Soil water storage and SWC were negatively correlated with f , with correlation coefficients of -0.77 and -0.75 , respectively. The high soil water storage and SWC in May, September and October were mainly due to low soil evaporation.

3.2. Vertical distribution of soil water lc-excess profile and water movement mechanism

During the observation period of 2018–2019, the lc-excess value of soil water showed similar profile distribution characteristics, namely,

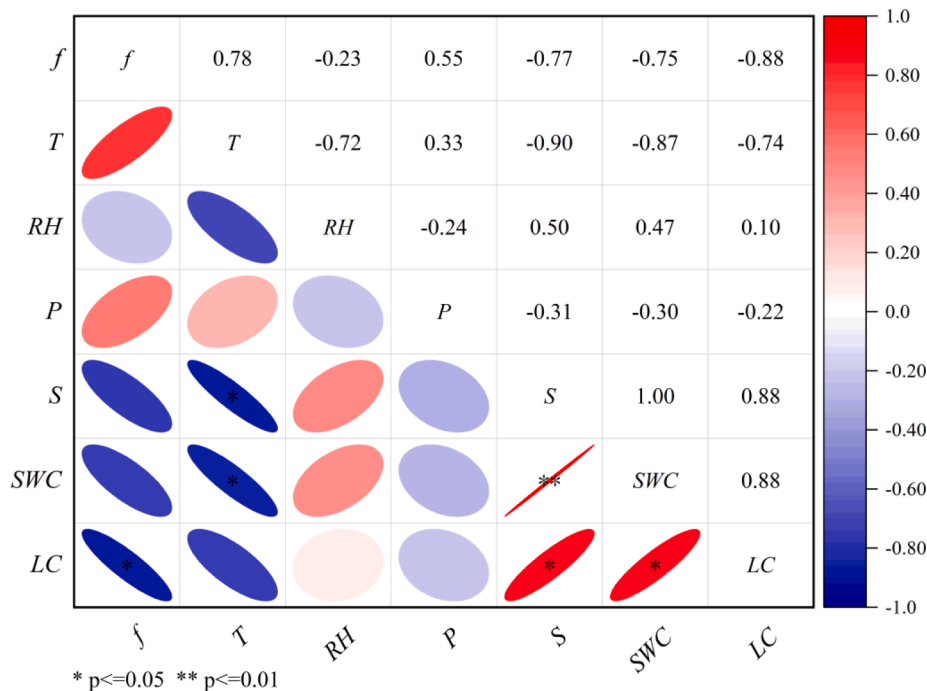


Fig.4. Correlation analysis diagram of soil evaporation loss rate f and other factors (Pearson correlation coefficient).

there was a large vertical change in 0–30 cm soil (except July 2018 and 2019), but the vertical change gradually decreased with the increase of soil depth, and showed a weak fluctuation or tended to be stable below 30–40 cm (except July 2018 and 2019). The lc-excess value of the precipitation isotope was 0‰, indicating that no evaporation has been experienced. The lc-excess value of water after evaporation was usually negative, and the lower the negative value was, the more intense the evaporation was. The vertical variation of lc-excess value in the sampled soil profile indicated the alternating variation in precipitation infiltration-soil evaporation. In July 2018, the soil water lc-excess value in the topmost layer of the plantation soil profile was -35.73‰ , indicating strong evaporation during the sampling period (Fig. 5c). In the 30–50 cm soil layer, soil water lc-excess was large and close to precipitation lc-excess (0‰), which represented that precipitation infiltration before sampling may be caused by the infiltration of the last large precipitation event. In the 60–100 cm soil layer, soil lc-excess began to be obviously negative, and was less than the corresponding precipitation value, indicating that the soil water isotope retained an obvious isotopic fractionation signal within the depth range of the soil layer. Similarly,

the infiltration signals of precipitation events were also displayed in the sampling profiles in August 2018 and July 2019. In August 2018, the soil water Lc-excess in the 20–100 cm soil layer was close to 0‰, which was mainly caused by the infiltration of multiple successive large-scale precipitation events before the sampling period in August. In July 2019, Lc-excess was also close to 0‰ in the 60–100 cm soil layer, which also indicated infiltration of precipitation event before sampling. Water isotope signals of soil water profiles showed high variability (Fig. 5a,b). At the same time, soil water storage at 80–100 cm depth showed a tendency of sudden increase (Fig. 5d), indicating that precipitation quickly flowed into the deep soils through the soil matrix by exposed soil cracks and roots.

3.3. Variation of soil water storage capacity of arid artificial forest

In this study, soil water was used to calculate the water storage in the 0–100 cm soil layer of the plantation during the 2018–2019 observation period. The average soil water storage of the plantations was 161.19 mm during the observation period, with the lowest at 0–10 cm (9.80 mm)

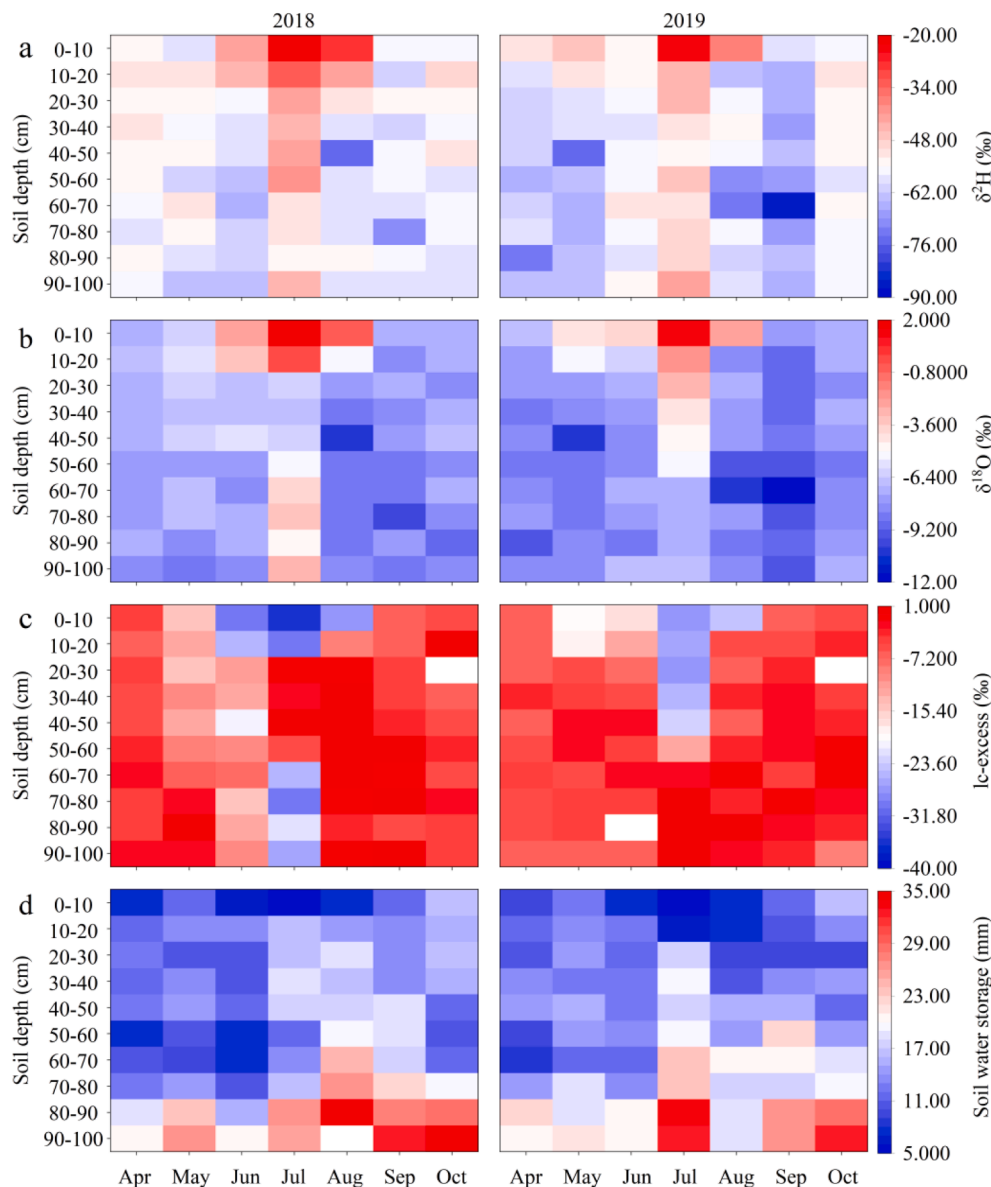


Fig. 5. Vertical distribution heat map of soil water $\delta^2\text{H}$ (a), $\delta^{18}\text{O}$ (b), lc-excess (c) and soil water storage (d) at a depth of 0–100 cm during the growth period of plantation.

and the highest at 90–100 cm (26.33 mm) (Fig. 5d). During the observation period, soil water storage in June, as the lowest in the year, were 114.76 mm in 2018 and 137.05 mm in 2019, respectively. In August 2018, soil water storage reached the highest in the observation period, at 214.09 mm, which showed a strong correlation with precipitation. However, from the perspective of the whole observation period, the water storage capacity in October was the highest (181.46 mm), followed by September (179.12 mm) (Fig. 6d). Besides September and October, the water storage of the 0–100 cm soil layer was also high in July and August. Although the temperature and soil evaporation loss rate were high in July and August, these two months also had the most precipitation (accounting for 60.25% of the total precipitation in the observation period). Therefore, the water storage was high.

To assess the soil water storage capacity of artificial forests in arid areas, we also calculated the water storage capacity of other vegetation types (grass and shrub) in the 0–100 cm soil layer during the observation period. The results showed that the water storage capacity of grass was slightly higher than that of forest and shrubland in the 0–40 cm soil layer; the water storage capacity of shrub was slightly higher than that of grassland and forestland in the 40–80 cm soil layer; the water storage capacity of forest was slightly higher than that of grassland and shrubland in 80–100 cm soil layer (Fig. 7 b,d). There was a significant difference of soil water storage between grasslands and plantations in the 20–70 cm soil layer, and also between shrublands and plantations in 50–70 cm soil layer (Fig. 7b,d). Based on the calculation of soil water storage, we further evaluated the soil water consumption rate of artificial forests compared with grassland and shrub in arid areas. The results showed that compared with grassland, the water consumption rate of the poplar plantation was 11.05 ± 3.25 mm/a; compared with shrubland, the water consumption rate of the poplar plantation was 7.37 ± 1.90 mm/a (Fig. 7 a,c).

4. Discussion

4.1. Mechanism of soil water movement in artificial forest in arid areas

The roots compresses the soil through interpenetration growth to form a root channel (Fan et al., 2017). This root channel can accelerate the horizontal and vertical water movement in the soil, thus improving the soil water infiltration rate (Ghestem et al., 2011; Zhang et al., 2017b). Precipitation enters into the soil rapidly through the root channel, which makes the precipitation have the characteristics of preferential flow recharge for soil water recharge in plantations (Fig. 5d). However, the root system also changes the spatial heterogeneity of soil structure (Zhang et al., 2017b). Due to soil heterogeneity, precipitation stayed in the soil for a long time, which made the newly replenished water fully mix with the original water (Zhu et al., 2022b). There was a significant difference of soil water between plantation and grassland at 20–70 cm, and a significant difference of soil water between plantation and shrubland at 50–70 cm. This shows that precipitation has different movement mechanism when passing through different soils such as artificial forest, grassland and shrubland (Fig. 8). Grassland roots are shallow and have not yet formed root channels (Ferrante et al., 2014), mainly characterized by plug flow replenishment. Shrubland roots is shallower than that of plantation, so the root channel of shrubland is shallower than that of plantation (Canadell et al., 1996). Precipitation replenishes the soil in the form of preferential flow in the 0–50 cm soil layer, and begins to switch to plug flow replenishment in 50–70 cm soil layer as the root channel begins to gradually shrink or even disappear. However, the artificial forest has a deeper root channel and water can quickly pass through the upper and middle layers of soil, so there is a significant difference in different soil layers compared with grassland and shrub (Fig. 7b,d).

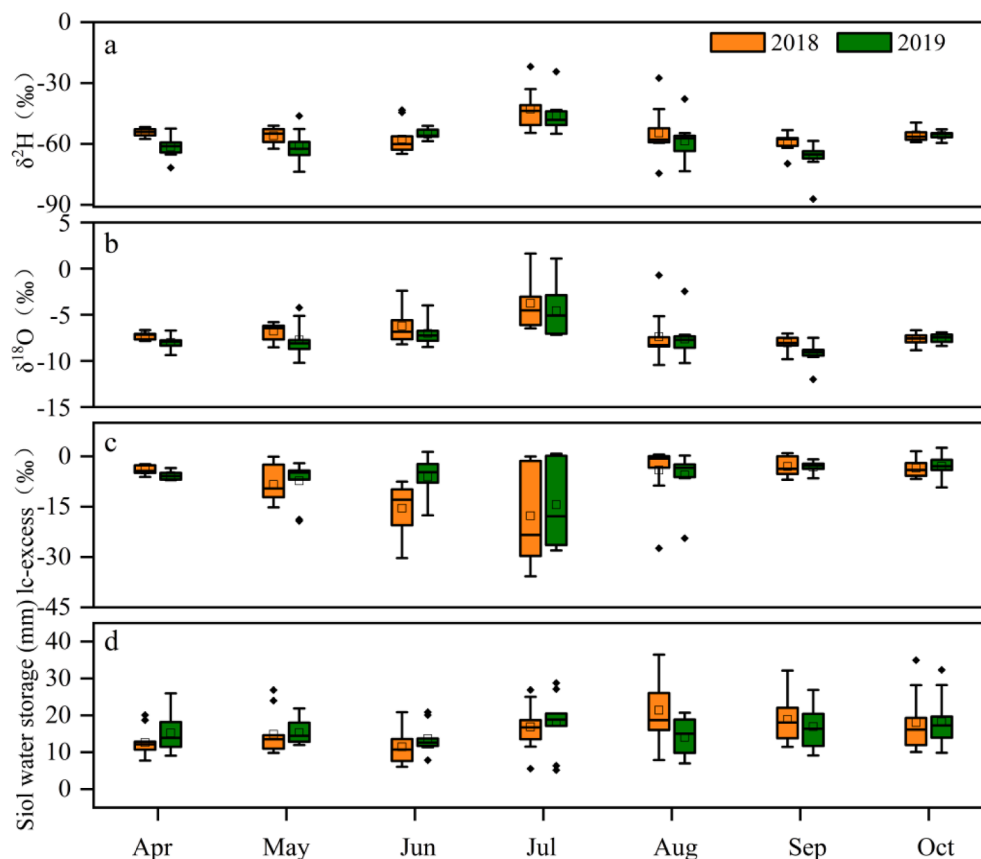


Fig. 6. Variation of soil water $\delta^2\text{H}$ (a), $\delta^{18}\text{O}$ (b), lc-excess (c) and soil water storage (d) during the plantation growth period (The horizontal line in the box diagram represents the median, and the square box represents the average).

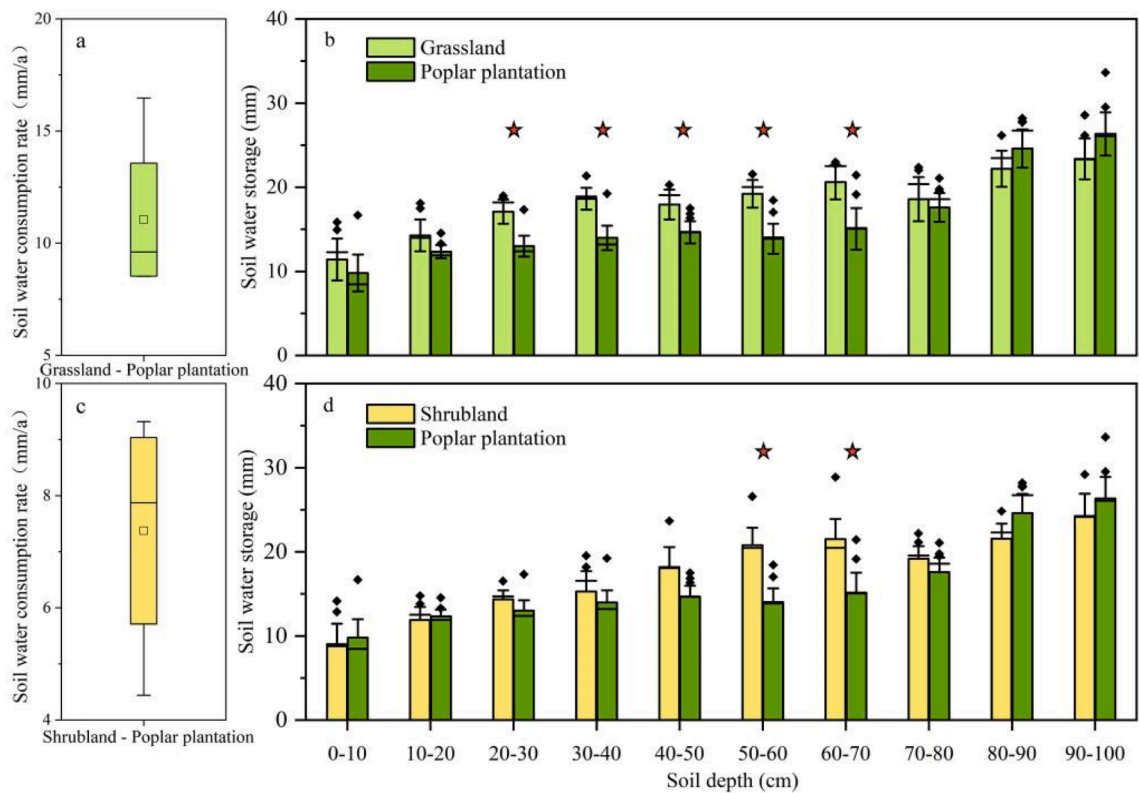


Fig. 7. Soil water storage and soil water consumption rate of grassland, shrub and poplar plantation (a, c. represent soil water consumption rate of poplar plantation in 2018 and 2019 respectively; b, d. represent the soil water storage comparison between poplar plantation and grassland, poplar plantation and shrubland at 0–100 cm in 2018 and 2019 respectively; the asterisk represents significant difference).

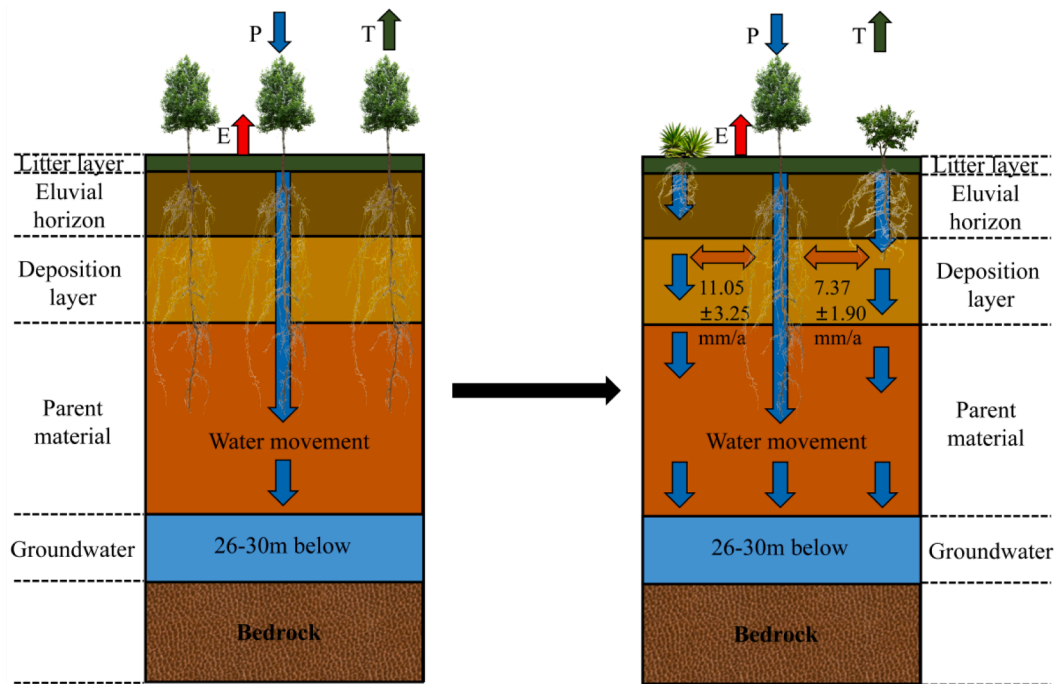


Fig. 8. Schematic diagram of soil water movement mechanism and water consumption rate of single poplar plantation and arbor (poplar)- shrub-grass plantation.

4.2. Evaluation of soil water storage capacity of artificial forest in arid areas

Soil water storage is an important indicator for evaluating ecosystem water resources conservation and water and soil conservation, which reflects the potential ability of soil to regulate water (Milly, 1993). The larger the water storage is, the greater the potential of soil to regulate rainfall redistribution (Xia et al., 2017). Grassland has the strongest water storage capacity in the surface soil, which is mainly due to the high density of grassland vegetation, the shallow distribution of root systems, and the developed root system in the surface soil, forming a tight root network (Ferrante et al., 2014). The combined effect of the grassland root system and aggregate soil structure can significantly increase the infiltration and storage of grassland soil on precipitation (He et al., 2012; Wu et al., 2016; Liu et al., 2019b). In addition, the soil organic matter (SOM) content of grassland soil is higher than that of forest and shrub, which is also an important reason for the high water storage capacity of grassland (Keen, 2011). The SOM is formed through the partial decomposition and transformation of plant litter by soil organisms (Cotrufo et al., 2015). Fragments produced during litter decomposition can promptly associate with the topsoil layer. The addition of organic matter to the soil increases porosity and decreases bulk density (Cotrufo et al., 2015; Liu et al., 2019a). So the organic matter layer can absorb a large amount of water, and at the same time, it can also better infiltrate water (Smith and Elliott, 1990). Litter crusts significantly increased SOM compared to biocrusts and bare lands, by 2.4 times and 3.8 times, in arid areas (Liu et al., 2019a). The poplar has more litter than the shrub (*Haloxylon ammodendron*, etc.), and the dead matter decomposes to supplement more humus to the soil surface layer than the shrub (Zhang et al., 2017a). Litter improves the soil structure of the top soil layer of arbour forest and improves the soil infiltration performance. Therefore, the surface water storage capacity of forests is also higher than that of shrubs (Shangguan and Zheng, 2006). In this study, the water storage capacity of plantation was the highest in October (Fig. 6). In addition to the low temperature, the main reason was that poplar leaves began to fall and form litter, which had a strong ability to absorb and store water. The water storage capacity of deep soils (80–100 cm) in the forest is much higher than that in shrubs and grass, which was mainly caused by the difference in vegetation roots. As the root system of trees is deeper and thicker than that of shrubs and grasses (Fig. 8), it extends, intersperses and loosens soil more strongly than that of shrubs and grasses, and the root channel formed is deeper (Canadell et al., 1996; Ferrante et al., 2014), which is conducive to more water entering deep soils.

4.3. Implications for artificial forest construction in arid areas

How to make better use of the limited water has become the main issue in the current ecological environment construction in the arid areas of northwest China, where the ecological environment is fragile, and precipitation is scarce (Song et al., 2017). Grassland, shrubland and forest all have certain water storage capacities. However, in arid areas where rainfall is scarce, shrubs and grass can grow luxuriantly, and form vegetation with large coverage and density, so the benefit of water and soil conservation is high (Wei et al., 2022); whereas, trees grow poorly and sparsely under dry soil conditions, with less litter, less coverage and thickness, so the benefit of water and soil conservation is lower (Cui et al., 2022). Moreover, the water consumption is higher than that of grassland and shrubland (Fig. 7a,c). This model of plantation construction greatly affects the water balance of the ecosystem.

Therefore, we must redesign the construction mode of artificial forests in arid areas to reduce soil water consumption and ensure the water balance of the ecosystem. In such designs, we can combine coniferous forest with broadleaved forest and pay equal attention to arbour, shrubs and grass (Chen et al., 2015). Under the condition that the density of the existing plantation remains unchanged, some trees can be replaced with

grass and shrubs, which can not only play the role of wind prevention and sand fixation, but also reduce soil water consumption by the large-scale poplar, so as to make full use of the advantages of grass and shrubs and make up for the weaknesses of poplars, so as to achieve the ideal effect (Fig. 8). At the same time, grass can better conserve surface soil water, and ensure the water for shrubs and trees growth, which greatly improved the growth rate of trees and promoted the growth of plant roots (Ferrante et al., 2014; Cui et al., 2019). The interpenetration, entanglement, network and consolidation of plant roots in the soil can make the soil loose and breathable, improve the soil porosity, and promote the infiltration and storage of soil water, which is of great significance to improve the soil water storage capacity (Jotisankasa and Sirirattanachai, 2017).

4.4. The limitations of the current study and potential improvements for future work

Soil water dissipation and movement are important parts in reflecting regional hydrological processes (Brooks et al., 2015). Current studies on soil moisture dynamics in arid areas using hydrogen oxygen stable isotope data mainly focus on oasis farmland (Yang et al., 2015; Wang et al., 2022). For example, researchers analyzed the infiltration process of irrigation water in Minqin oasis farmland (maize field) and estimated the proportion of irrigation water contributed to soil water in different soil layers and the effective contribution time by using stable isotopes of soil water in farmland (Zhu et al., 2021b). However, there is insufficient refined soil and water isotope observations in natural/semi-natural landscapes such as grasslands, shrubs and artificial forests. In this study, the soil evaporation loss rate and water storage capacity of poplar plantation in arid areas were quantified by using stable isotope data, and the movement mechanism of soil water was revealed. However, there are some limitations in this study. Due to the lack of detailed observations, such as tracking and sampling of a single precipitation event, it is difficult to quantify the contribution proportion and time of a single precipitation event to soil water. Meanwhile, since the soil profile sampling depth (1 m) does not cover the complete root depth of poplar plantation (Xi et al., 2016), the effect of plantation construction on the consumption and movement mechanism of deeper soil water in arid areas has not been further studied. In the future research on soil water dynamics of plantations in arid areas, we should further strengthen the sampling and observation of soil water isotope data. Moreover, we should further strengthen the process research by which artificial forests in arid areas alter the regional water cycle and affect the components of regional water balance (Schwäzel et al., 2020). This will help us to better understand plantation hydrological regulating function and the influence of vegetation type on soil water dynamics (Newman et al., 2006).

5. Conclusion

In this study, we used stable isotopes to investigate the evaporation, movement and storage of soil water in the artificial forest in an arid oasis area, and provided further insights into the soil water dissipation and movement in an artificial forest. The results showed that the average soil evaporation loss rate in the plantation was 21.30%, with the highest evaporation loss rate in summer. High temperature and sufficient precipitation were the main driving forces of soil water evaporation loss in the plantation. The recharge of precipitation to plantation soil water has the characteristics of preferential flow recharge. This may be because the roots compress the soil through interpenetration growth to form a root channel, enabling precipitation to quickly pass through the soil matrix. Due to soil heterogeneity, precipitation stayed in the soil for a long time, which made the newly recharged water fully mix with the original water. The average soil water storage capacity of 0–100 cm in the plantation was 161.19 mm, and the water storage capacity of 80–100 cm in the plantation was slightly stronger than that in grassland

and shrub. However, the water consumption rate of the plantation was higher than that of grassland and shrubland, and compared with grassland and shrubland, the water consumption rate of the plantation was 11.05 ± 3.25 mm/a and 7.37 ± 1.90 mm/a, respectively. Although constructing artificial forests in arid areas can increase the deep soil water storage on a certain basis, it will also consume more soil water. Therefore, we believe that the combination of arbor, shrub and grass planting in arid areas can better conserve water, reduce soil water consumption and ensure the growth of artificial forests. This study mainly emphasized the temporal and spatial heterogeneity of soil water evaporation, infiltration and storage in plantations. These results are very important for understanding the regional hydrological process and ecological restoration of artificial forests in arid areas and coordinating ecological benefits, landscape benefits and water resource utilization benefits.

CRedit authorship contribution statement

Dongdong Qiu: Writing – original draft. **Guofeng Zhu:** Writing – review & editing. **Xinrui Lin:** Conceptualization. **Yinying Jiao:** Validation. **Siyu Lu:** Methodology. **Jiatong Liu:** Validation. **Jiawei Liu:** Conceptualization. **Wenhao Zhang:** Data curation. **Linlin Ye:** Supervision. **Rui Li:** Software. **Qinqin Wang:** Investigation. **Longhu Chen:** Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author.

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